

## 5.Diabetes Dataset — Linear Regression

The PDF asks you to analyze diabetes.csv using **linear regression and multiple regression in WEKA**.

### Software:WEKA

#### Steps

1. Open **WEKA** → **Explorer**.
2. Click **Preprocess**.
3. Click **Open file**.
4. Select:

diabetes.csv

5. Check that the attributes are loaded correctly.
6. Click **Classify**.
7. Click **Choose**.
8. Select:

Functions → LinearRegression

9. Select the required numeric target/class attribute.
10. Select **10-fold cross-validation**.
11. Click **Start**.
12. Record the regression equation and error values.
13. Take a screenshot.

### Programming Code : diabetes.csv

#### OUTPUT :

The screenshot displays the WEKA Explorer window with the 'Classify' tab selected. The 'Classifier' dropdown is set to 'LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4'. Under 'Test options', 'Cross-validation' is selected with 'Folds' set to 10. The '(Num) Outcome' dropdown is set to 'Outcome'. The 'Start' button has been clicked, and the 'Result list' on the left shows '09:40:50 - functions.LinearRegression' selected. The main output area displays the following information:

```
Classifier output
Attributes: 9
Pregnancies
Glucose
BloodPressure
SkinThickness
Insulin
BMI
DiabetesPedigreeFunction
Age
Outcome
Test mode: 10-fold cross-validation

=== Classifier model (full training set) ===

Linear Regression Model

Outcome =

0.0209 * Pregnancies +
0.0057 * Glucose +
-0.0024 * BloodPressure +
0.0131 * BMI +
0.1403 * DiabetesPedigreeFunction +
0.0028 * Age +
-0.8263

Time taken to build model: 0.09 seconds

=== Cross-validation ===
=== Summary ===

Correlation coefficient      0.5322
Mean absolute error         0.3366
Root mean squared error     0.4036
Relative absolute error     74.0119 %
Root relative squared error 84.6013 %
Total Number of Instances   768
```

The status bar at the bottom shows 'OK' and a 'Log' button.